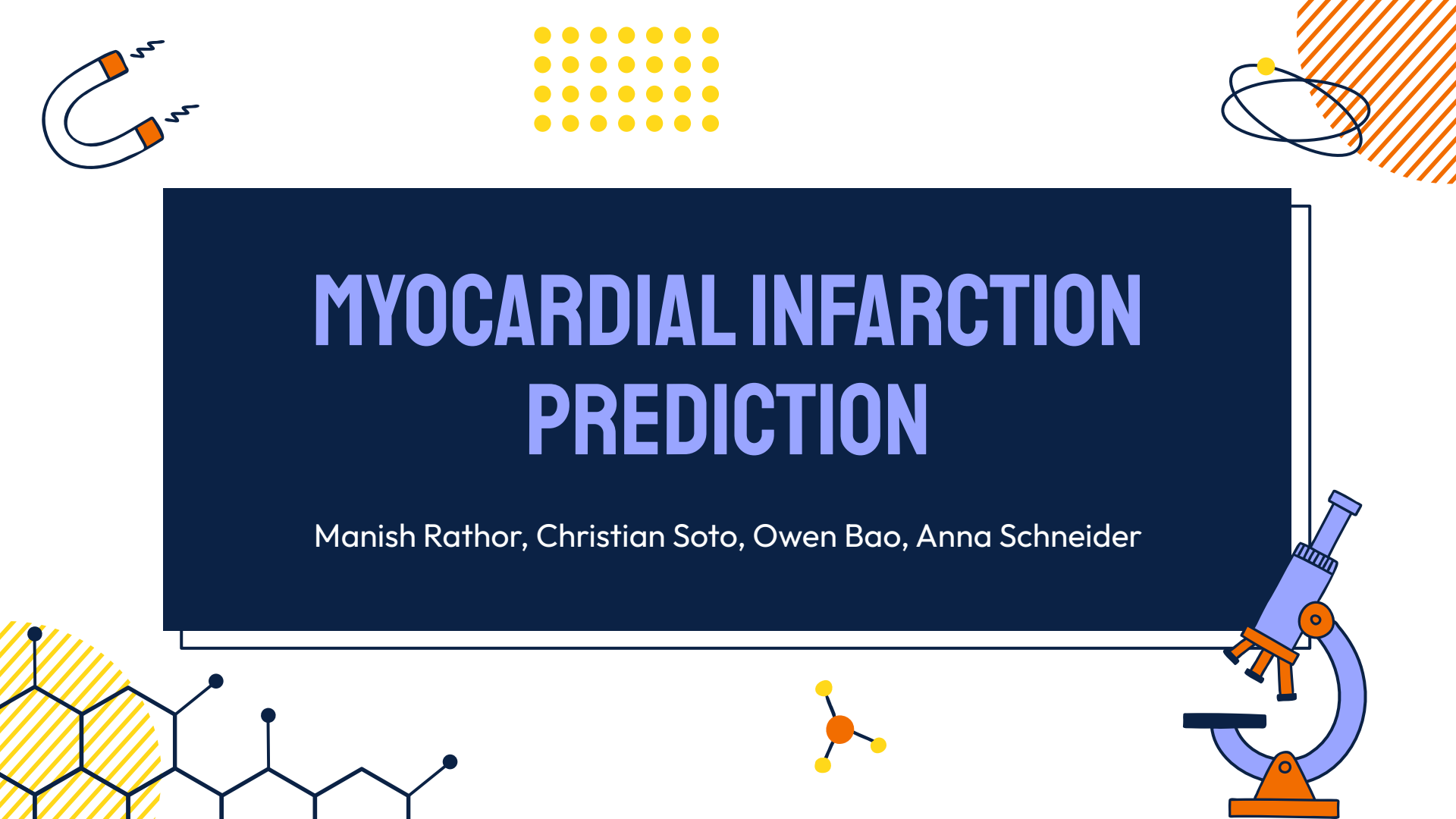




MYOCARDIAL INFARCTION PREDICTION

Manish Rathor, Christian Soto, Owen Bao, Anna Schneider



TABLE OF CONTENTS



01

INTRODUCTION

Question of Interest



02

DATA ANALYSIS

Exploratory Analysis +
Preprocessing

03

STATISTICAL ANALYSIS

Dimensionality Reduction
+ Statistical Learning

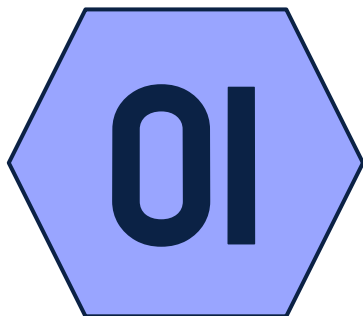
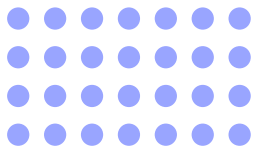


04

CONCLUSION

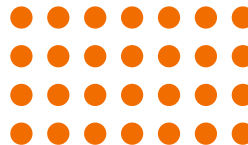
Analysis of Results +
Future Implications



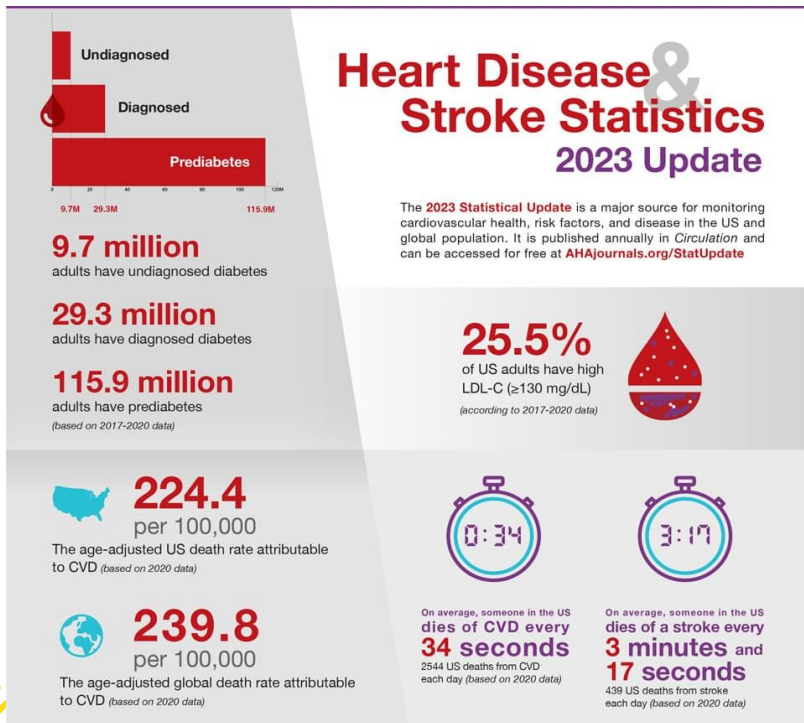


INTRODUCTION

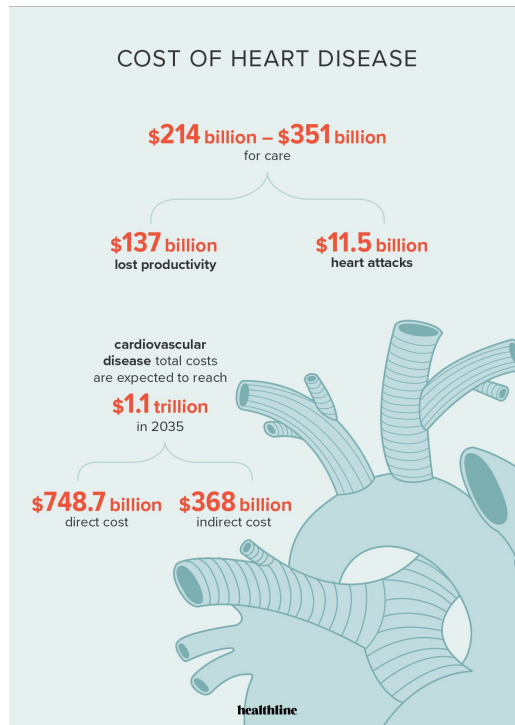
What are we investigating?



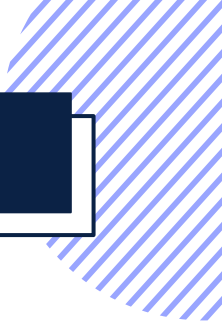
HEART DISEASE IN THE UNITED STATES



American Heart Association

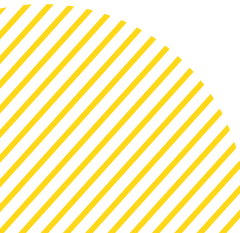


Healthline (via CDC)



OUR QUESTION

Can we develop a **method** to
identify a patient's risk of
heart-related death?



THE DATASET

TITLE

Myocardial Infarction
Complications

DATA TYPES

Categorical: 22
Binary: 79
Continuous: 11

DIMENSIONS

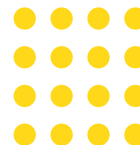
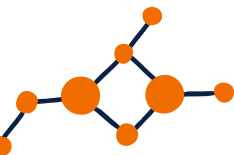
Rows: 1700
Features: 111

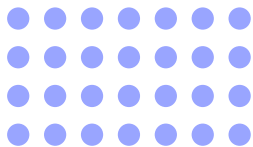
TARGET

Lethal Outcome

MISSING VALUES

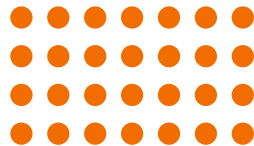
15974





DATA ANALYSIS

Exploring and Cleaning our Dataset



COMPONENTS



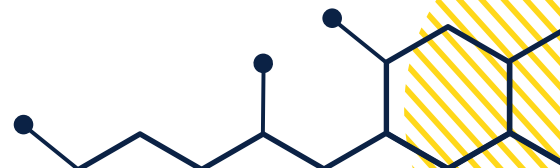
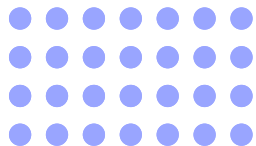
PREPROCESSING

Missing Value Imputation



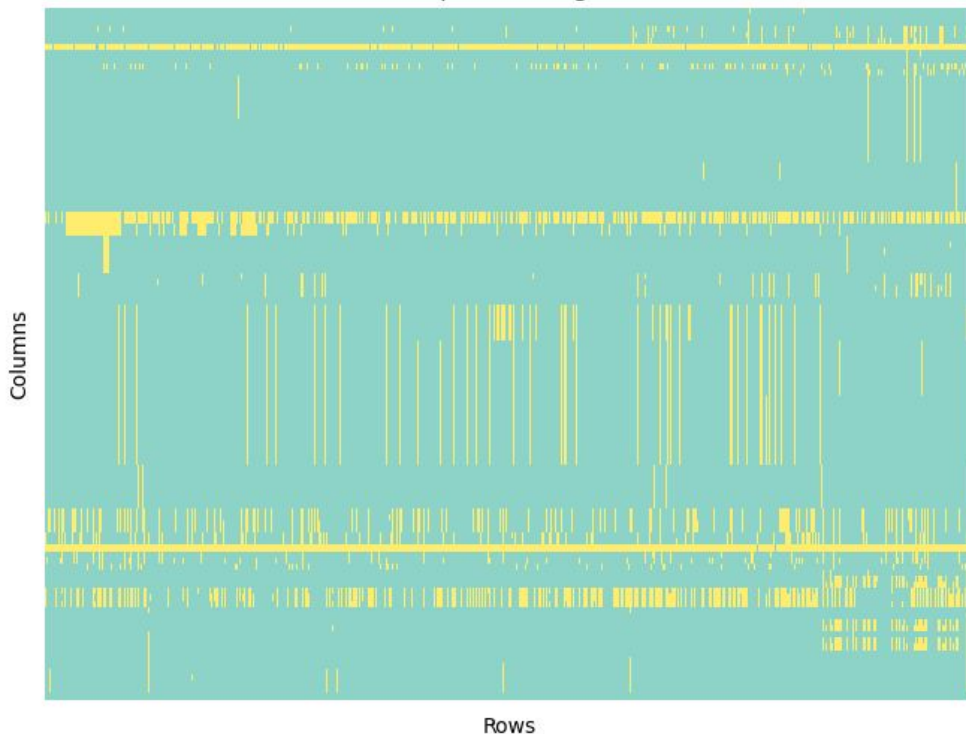
EXPLORATORY ANALYSIS

Data Visualization +
Correlation/Pattern Analysis



MISSING VALUES

Heatmap of Missing Values



IMMEDIATE SOLUTION

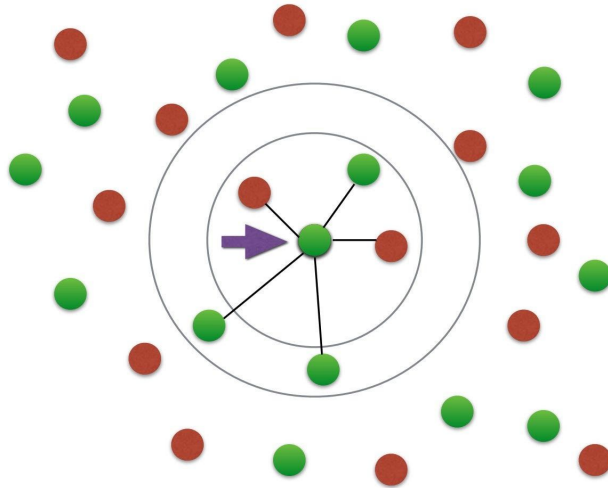
Removed Rows and Columns
with 25% or more of data
missing

NEXT STEP

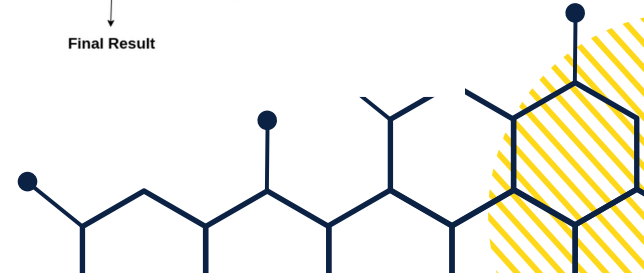
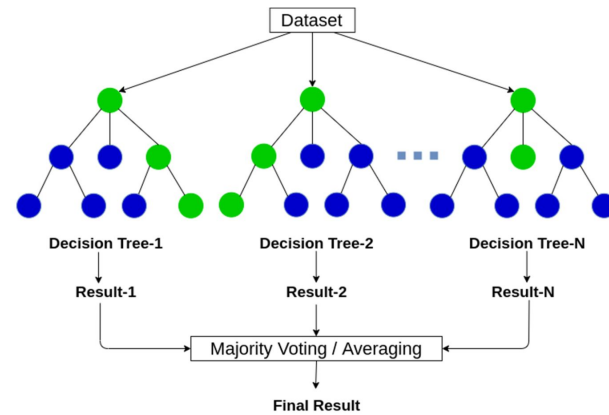
Reverse
Regression/Classification for
Imputation

IMPUTATION METHODS

K-Nearest Neighbours Imputation



Iterative Imputation w/ Random Forest



COMPARING METHODS

ITERATIVE IMPUTER

PROS

Allows for User Control

Allows for Various Methods

CONS

Computationally Intensive

Requires In-Depth Parameter Tuning

KNN IMPUTER

PROS

Computationally Rapid

Restricted to one Method

CONS

Not Effective for Binary Data

Minimal Parameter Tuning

RESULTS:

Approximately 75% of the imputations matched

OPTIMAL SOLUTION:

Iterative Imputer w/ Random Forest

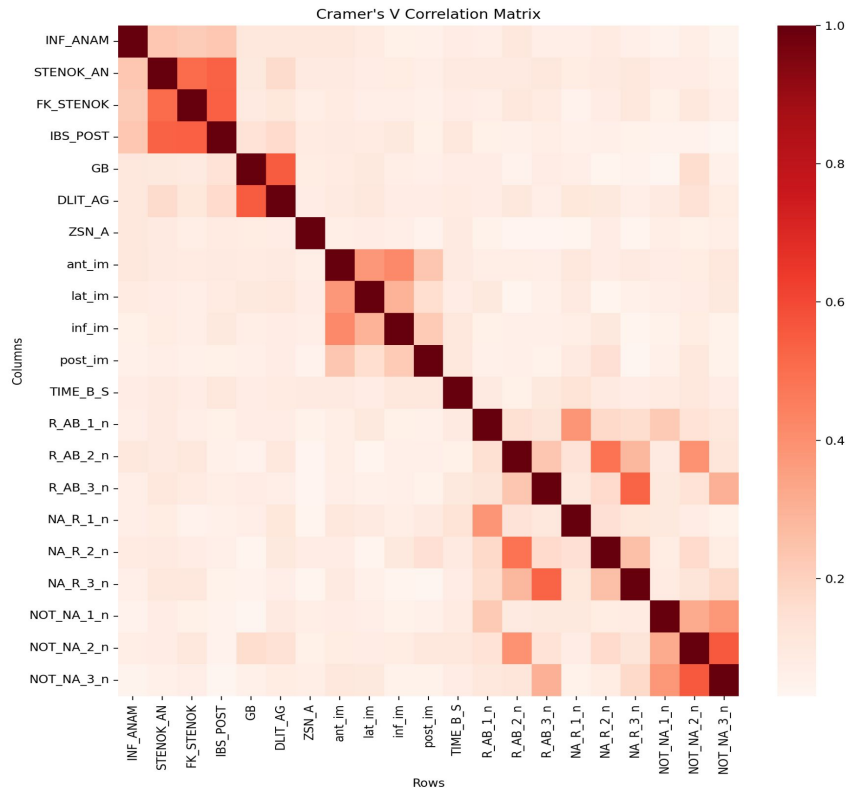


Measures Association between Two Binary Variables

	y = 0	y = 1
x = 0	A	B
x = 1	C	D

$$\phi = \frac{ad - bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}$$

CORRELATION ANALYSIS - CATEGORICAL DATA



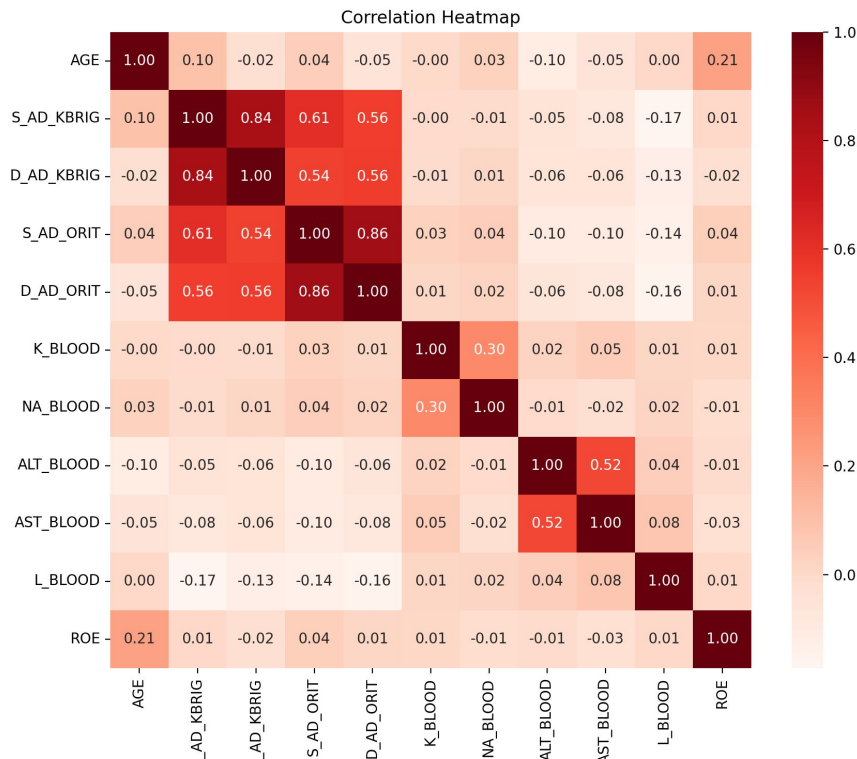
Cramer's V

Measures Association between
Two Categorical Variables

	Eye Color		
	Blue	Green	Brown
Republican	8	5	6
Democrat	2	8	3
Independent	4	6	8

$$V = \sqrt{\frac{\chi^2}{N(k-1)}}$$

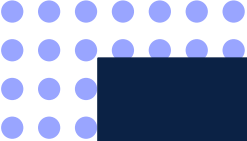
CORRELATION ANALYSIS - CONTINUOUS DATA



Pearson Correlation Coefficient

Measures correlation between continuous variables

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$



CORRELATION ANALYSES FINDINGS



BINARY

Variables representing heartbeat-related measurements were positively correlated

Mutually exclusive sets of binary variables demonstrated negative correlation

CATEGORICAL

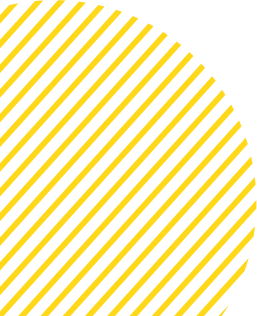
Features related to a patient's history of heart disease were positively correlated

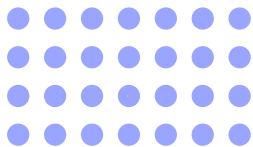
Variables documenting opioid usage were positively correlated with data documenting pain relapse

CONTINUOUS

Blood-pressure related data demonstrated high correlation

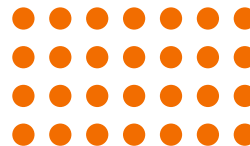
Blood-AIAT and Blood-AsAT content had high positive correlation (Liver health biomarkers)





STATISTICAL ANALYSIS

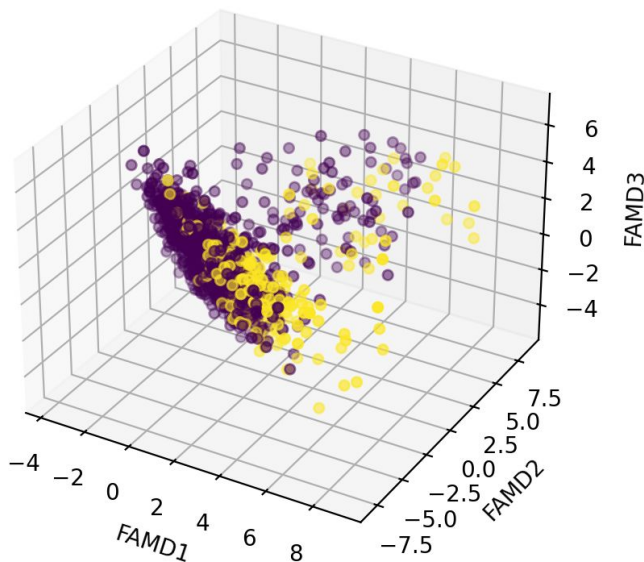
Reducing the Dataset + Predictions



FACTOR ANALYSIS OF MIXED DATA

Dimensionality Reduction technique
for Mixed Data Sets

“Combines” PCA and MCA

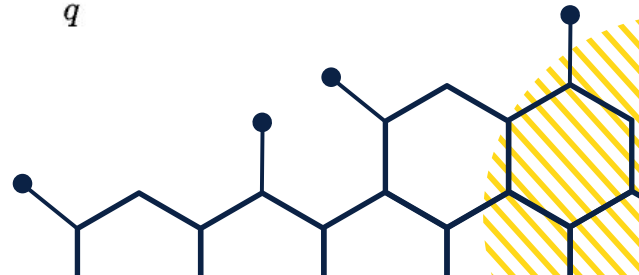


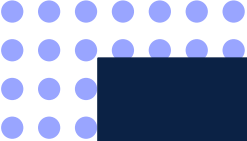
We have K quantitative variables $1, \dots, K$ and
 Q qualitative variables $1, \dots, Q$.

z is a quantitative variable.

- $r(z, k)$ is the correlation coefficient between variables k and z
- $\eta^2(z, q)$ is the squared correlation ratio between variables z and q

$$\sum_k r^2(z, k) + \sum_q \eta^2(z, q) \text{ maximum.}$$



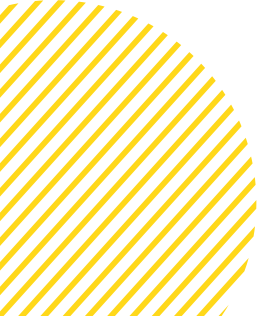


COMPONENT FINDINGS



COMPONENT 1

Influenced most by factors
regarding heart rate and
chest pain



COMPONENT 2

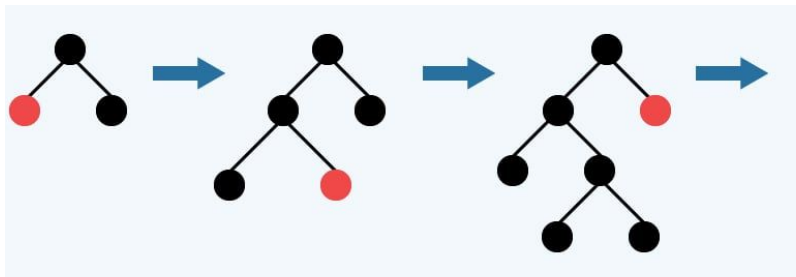
Influenced most by factors
regarding heart rate

COMPONENT 3

Influenced most by
blood-electrolyte levels

STATISTICAL LEARNING METHODOLOGY

Gradient Boosted Model



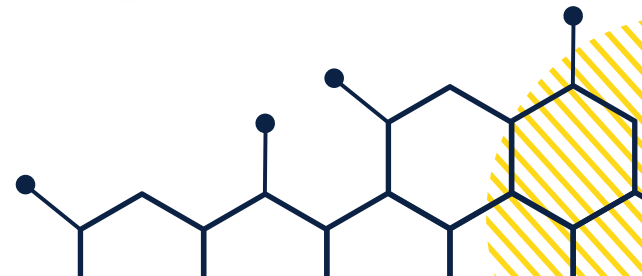
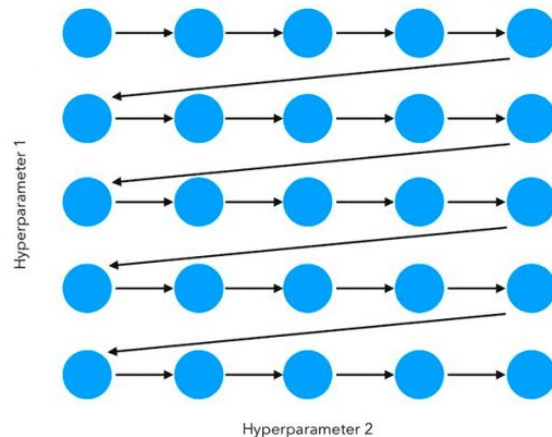
Best Estimator

Number of Estimators: 300

Learning Rate: 0.01

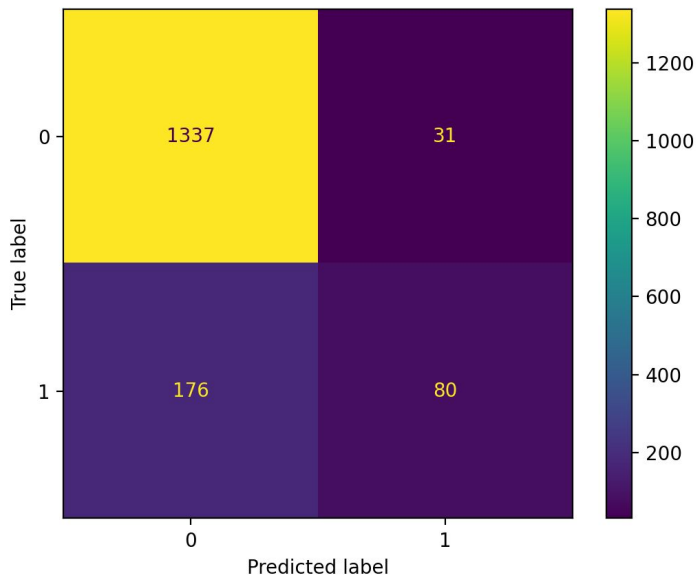
Maximum Depth: 2

Grid Search



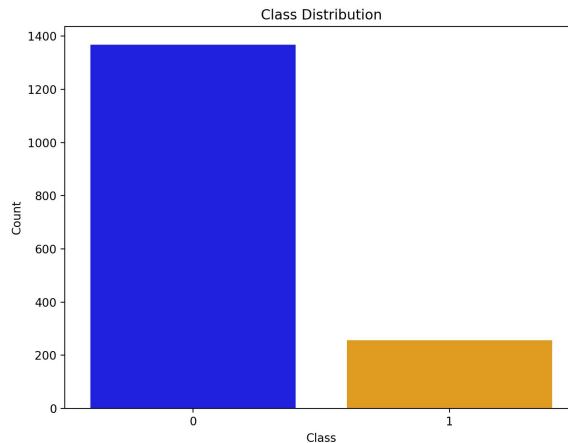
MODEL PERFORMANCE

Confusion Matrix

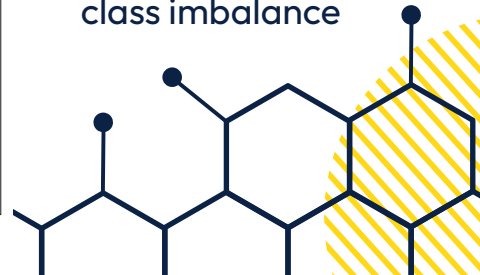


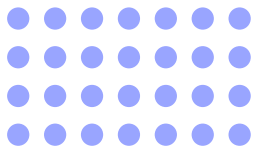
Classification Report

	precision	recall	f1-score	support
0	0.88	0.98	0.93	1368
1	0.72	0.31	0.44	256
accuracy			0.87	1624
macro avg	0.80	0.64	0.68	1624
weighted avg	0.86	0.87	0.85	1624



Poor
performance
can be partially
attributed to
class imbalance

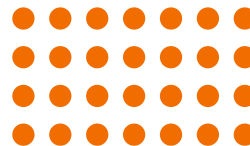




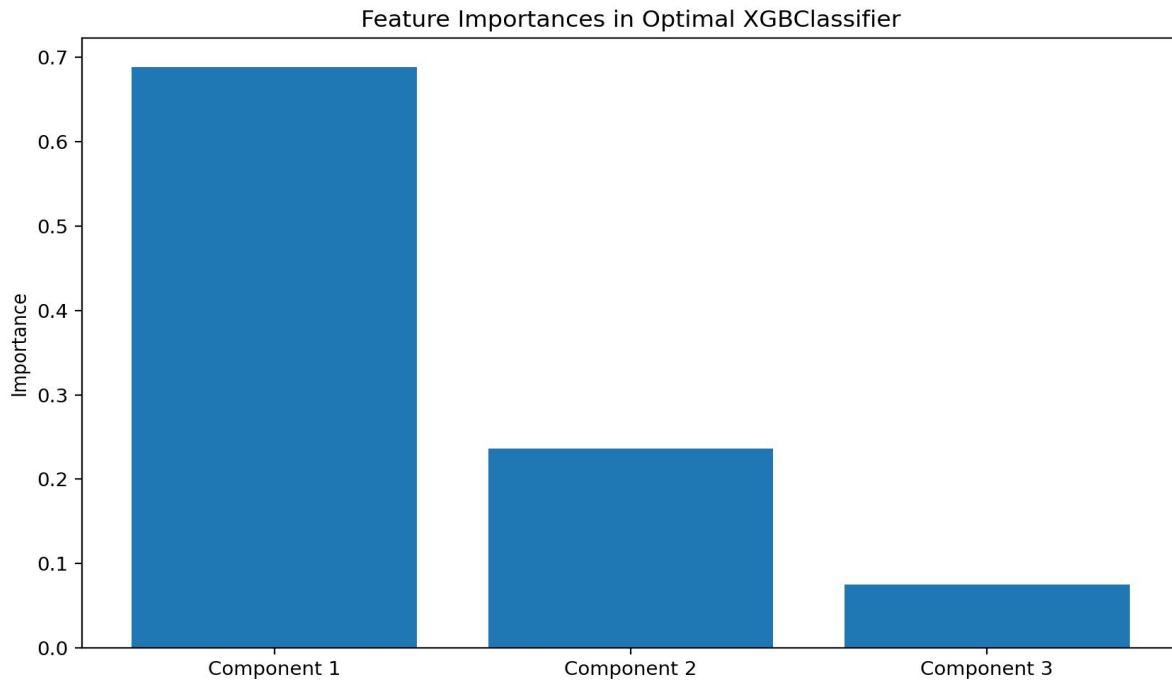
04

CONCLUSION

Takeaways + Looking Forward

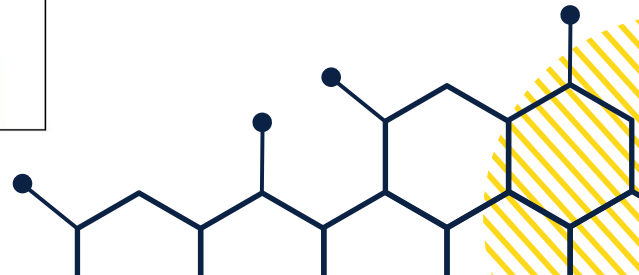


FEATURE IMPORTANCE



Predictions are driven
by patients' heart rate
and chest pain levels

Blood-Electrolyte
levels are less
influential





STRENGTHS AND WEAKNESSES

STRENGTHS

Effective Dimensionality Reduction

Strong Predictive Performance for Surviving Patients

WEAKNESSES

Minimal Analysis of Individual Features

Class Imbalance led to Poor Predictive Performance for Lethal Outcome

FUTURE IMPROVEMENTS

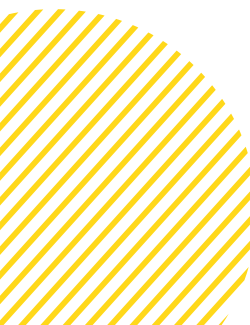
Pattern Analysis/Data Visualization to find insights within features

Testing more methods to account for temporal influence

Finding more data to improve class imbalance

Leveraging the correlation analyses findings for feature selection





THANK YOU

